A2

Q1.1: Decision Trees

Question 1~5

Question 1	Question 2	Question 3	Question 4	Question 5
A	AC	C	A	A

Writing Part

Part 1: Calculating Information Gain and Selecting the Optimal Splitting Attribute

Information gain measures how much uncertainty (entropy) in the target variable ("Play?") is reduced by splitting the dataset on a particular attribute. It is calculated as:

$$\text{Information Gain} = \text{Entropy}_{\text{before}} - \text{Entropy}_{\text{after}}$$

where:

- $\text{Entropy}_{\text{before}}$ is the entropy of the entire dataset.
- $\text{Entropy}_{\text{after}}$ is the weighted average entropy of the subsets after splitting on an attribute.

The entropy for a set with two classes (e.g., "Yes" and "No") is:

$$\text{Entropy} = -p_{\text{yes}} \log_2(p_{\text{yes}}) - p_{\text{no}} \log_2(p_{\text{no}})$$

where p_{yes} and p_{no} are the proportions of "Yes" and "No" instances.

Step 1: Entropy of the Entire Dataset

The dataset has 10 instances:

- "Yes": Days 2, 3, 4, 7, 9 (5 instances)
- "No": Days 1, 5, 6, 8, 10 (5 instances)

Proportions:

- $p_{\text{yes}} = 5/10 = 0.5$
- $p_{\text{no}} = 5/10 = 0.5$

Entropy:

$$\text{Entropy}_{\text{before}} = -0.5 \log_2(0.5) - 0.5 \log_2(0.5)$$

$$\log_2(0.5) = -1$$

$$\text{Entropy}_{\text{before}} = -0.5 \times (-1) - 0.5 \times (-1) = 0.5 + 0.5 = 1$$

So, the initial entropy is 1.

Step 2: Information Gain for Each Attribute

Weather (Sunny, Cloudy, Rainy)

- **Sunny**: Days 1, 3, 8 (3 instances)
  - Yes: 1 (Day 3)
  - No: 2 (Days 1, 8)
  - $p_{\text{yes}} = \frac{1}{3}$, $p_{\text{no}} = \frac{2}{3}$
  - Entropy = $-\frac{1}{3}\log_2(\frac{1}{3}) - \frac{2}{3}\log_2(\frac{2}{3}) \approx 0.918$
- **Cloudy**: Days 2, 4, 9 (3 instances)
  - Yes: 3

- No: 0
- Entropy = $-1\log_2(1) - 0\log_2(0) = 0$ (pure subset)

- **Rainy**: Days 5, 6, 7, 10 (4 instances)
  - Yes: 1 (Day 7)
  - No: 3 (Days 5, 6, 10)
  - $p_{\text{yes}} = \frac{1}{4}$, $p_{\text{no}} = \frac{3}{4}$
  - Entropy = $-\frac{1}{4}\log_2(\frac{1}{4}) - \frac{3}{4}\log_2(\frac{3}{4}) \approx 0.811$

Weighted entropy:

$$\text{Entropy}_{\text{after}} = \frac{3}{10} \times 0.918 + \frac{3}{10} \times 0 + \frac{4}{10} \times 0.811 = 0.5998$$

Information Gain:

$$1 - 0.5998 = 0.4002$$

Temperature (Hot, Mild, Cool)

- **Hot**: Days 1, 2, 8, 9 (4 instances)
  - Yes: 2 (Days 2, 9)
  - No: 2 (Days 1, 8)
  - Entropy = $-0.5\log_2(0.5) - 0.5\log_2(0.5) = 1$
- **Mild**: Days 3, 4, 5, 7, 10 (5 instances)
  - Yes: 3 (Days 3, 4, 7)
  - No: 2 (Days 5, 10)
  - $p_{\text{yes}} = 3/5 = 0.6$, $p_{\text{no}} = 0.4$
  - Entropy = $-0.6\log_2(0.6) - 0.4\log_2(0.4) \approx 0.971$
- **Cool**: Day 6 (1 instance)
  - Yes: 0
  - No: 1
  - Entropy = 0 (pure)

Weighted entropy:

$$\text{Entropy}_{\text{after}} = \frac{4}{10} \times 1 + \frac{5}{10} \times 0.971 + \frac{1}{10} \times 0 = 0.8855$$

Information Gain:

$$1 - 0.8855 = 0.1145$$

Humidity (High, Normal)

- **High**: Days 1, 2, 4, 5, 7, 8, 10 (7 instances)
  - Yes: 3 (Days 2, 4, 7)
  - No: 4 (Days 1, 5, 8, 10)
  - $p_{\text{yes}} = \frac{3}{7}$, $p_{\text{no}} = \frac{4}{7}$
  - Entropy = $-\frac{3}{7}\log_2(\frac{3}{7}) - \frac{4}{7}\log_2(\frac{4}{7}) \approx 0.985$
- **Normal**: Days 3, 6, 9 (3 instances)
  - Yes: 2 (Days 3, 9)
  - No: 1 (Day 6)
  - $p_{\text{yes}} = \frac{2}{3}$, $p_{\text{no}} = \frac{1}{3}$
  - Entropy = $-\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3}) \approx 0.918$

Weighted entropy:

$$\text{Entropy}_{\text{after}} = \frac{7}{10} \times 0.985 + \frac{3}{10} \times 0.918 = 0.9649$$

Information Gain:

$$1 - 0.9649 = 0.0351$$

Wind (Weak, Strong)

- **Weak**: Days 1, 2, 7, 9 (4 instances)
  - Yes: 3 (Days 2, 7, 9)

- No: 1 (Day 1)
- $p_{\text{yes}} = \frac{3}{4}$, $p_{\text{no}} = \frac{1}{4}$
- Entropy = $-\frac{3}{4}\log_2(\frac{3}{4}) - \frac{1}{4}\log_2(\frac{1}{4}) \approx 0.811$

- **Strong**: Days 3, 4, 5, 6, 8, 10 (6 instances)
  - Yes: 2 (Days 3, 4)
  - No: 4 (Days 5, 6, 8, 10)
  - $p_{\text{yes}} = \frac{1}{3}$, $p_{\text{no}} = \frac{2}{3}$
  - Entropy ≈ 0.918

Weighted entropy:

$$\text{Entropy}_{\text{after}} = \frac{4}{10} \times 0.811 + \frac{6}{10} \times 0.918 = 0.8752$$

Information Gain:

$$1 - 0.8752 = 0.1248$$

Step 3: Select Optimal Splitting Attribute

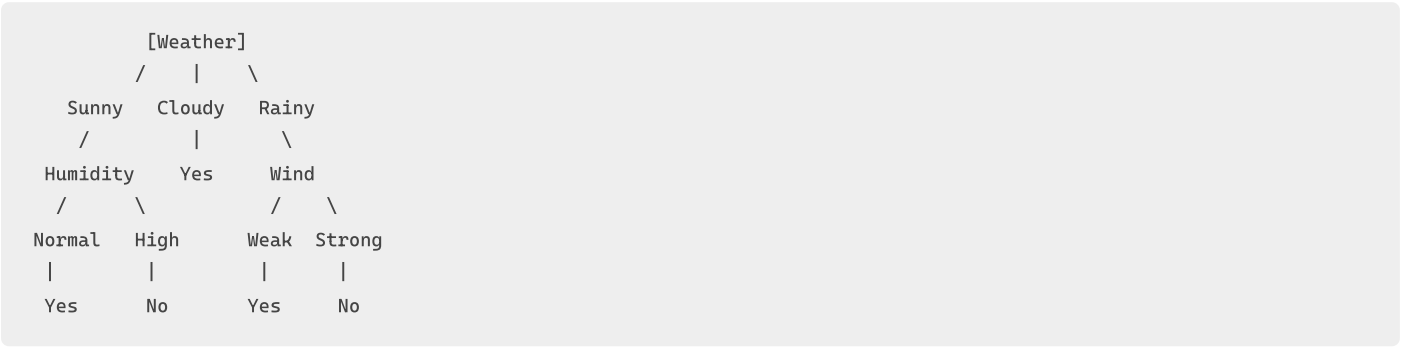
- Weather: 0.4002
- Temperature: 0.1145
- Humidity: 0.0351
- Wind: 0.1248

The highest information gain is for **Weather (0.4002)**, so it is the optimal splitting attribute for the first layer.

Part 2: Final Tree Structure

Starting with Weather as the root, build the tree as follows:

- **Weather = Cloudy**: Days 2, 4, 9 → All "Yes" (pure, leaf node: Play = Yes)
- **Weather = Sunny**: Days 1, 3, 8 (1 Yes, 2 No)
  - Split on **Humidity**:
    - **High**: Days 1, 8 → All "No" (leaf: Play = No)
    - **Normal**: Day 3 → "Yes" (leaf: Play = Yes)
- **Weather = Rainy**: Days 5, 6, 7, 10 (1 Yes, 3 No)
  - Split on **Wind**:
    - **Weak**: Day 7 → "Yes" (leaf: Play = Yes)
    - **Strong**: Days 5, 6, 10 → All "No" (leaf: Play = No)



Q1.2: CNN

For convolutional layers, the output spatial dimensions are calculated using:

$$W_{\text{out}} = \left\lfloor \frac{W_{\text{in}} + 2P - K}{S} \right\rfloor + 1$$

where W_{in} is the input width (or height), P is padding, K is the filter size, and S is the stride. For pooling layers, the formula is similar, adjusting for the lack of padding where specified. Parameters include weights and biases for convolutional and fully-connected layers, while pooling layers have none.

Layer 1: Input Layer

- **Activation Shape**: Given as $28 \times 28 \times 1$, representing a 28x28 grayscale image with 1 channel.
- **Parameters**: The input layer has no trainable parameters.
- **Result**:
  - Activation Shape: $(28, 28, 1)$
  - **Parameters: 0**

Layer 2: Conv4(15)

- **Input Shape**: $(28, 28, 1)$
- **Specifications**: 15 filters of size $4 \times 4 \times D$, where $D = 1$ (input depth), stride = 1, padding = 1.
- **Activation Shape**:
  - Width: $\left\lfloor \frac{28+2 \cdot 1-4}{1} \right\rfloor + 1 = 27$
  - Height: Similarly, 27 (since input is square).
  - Depth: 15 (number of filters).
  - Output Shape: $(27, 27, 15)$
- **Parameters**:
  - Each filter: $4 \times 4 \times 1 = 16$ weights + 1 bias = 17 parameters.
  - Total: $15 \times 17 = 255$.
- **Result**:
  - Activation Shape: $(27, 27, 15)$
  - **Parameters: 255**

Layer 3: Maxpool_2

- **Input Shape**: $(27, 27, 15)$
- **Specifications**: 2×2 filter, stride = 2, padding = 0.
- **Activation Shape**:
  - Width: $\left\lfloor \frac{27-2}{2} \right\rfloor + 1 = 13$
  - Height: Similarly, 13.
  - Depth: 15 (unchanged).
  - Output Shape: $(13, 13, 15)$
- **Parameters**: Pooling layers have no trainable parameters.
- **Result**:
  - Activation Shape: $(13, 13, 15)$
  - **Parameters: 0**

Layer 4: Conv2(25)

- **Input Shape**: $(13, 13, 15)$
- **Specifications**: 25 filters of size $2 \times 2 \times D$, where $D = 15$ (previous depth), stride = 1, padding = 0.
- **Activation Shape**:
  - Width: $\left\lfloor \frac{13+2 \cdot 0-2}{1} \right\rfloor + 1 = 12$
  - Height: Similarly, 12.
  - Depth: 25 (number of filters).
  - Output Shape: $(12, 12, 25)$
- **Parameters**:
  - Each filter: $2 \times 2 \times 15 = 60$ weights + 1 bias = 61 parameters.
  - Total: $25 \times 61 = 1525$.
- **Result**:
  - Activation Shape: $(12, 12, 25)$
  - **Parameters: 1525**

Layer 5: Maxpool_2

- **Input Shape**: $(12, 12, 25)$
- **Specifications**: 2×2 filter, stride = 2, padding = 0.
- **Activation Shape**:
  - Width: $\left\lfloor \frac{12-2}{2} \right\rfloor + 1 = 6$
  - Height: Similarly, 6.
  - Depth: 25 (unchanged).
  - Output Shape: $(6, 6, 25)$
- **Parameters**: No trainable parameters.
- **Result**:
  - Activation Shape: $(6, 6, 25)$
  - **Parameters: 0**

Layer 6: FC5

- **Input Shape**: $(6, 6, 25)$
- **Specifications**: Fully-connected layer with 5 output neurons. The input is flattened to $6 \times 6 \times 25 = 900$ neurons.
- **Activation Shape**: Output is 5 neurons, typically represented as $(5,)$ for a single sample.
- **Parameters**:
  - Each neuron: 900 weights (one per input neuron) + 1 bias.
  - Total: $5 \times (900 + 1) = 4505$.
- **Result**:
  - Activation Shape: $(5,)$
  - **Parameters: 4505**

Final Table

NO.	Layer	Activation Shape	# Parameters
1	Input Layer	(28, 28, 1)	0
2	Conv4(15)	(27, 27, 15)	255
3	Maxpool2	(13, 13, 15)	0
4	Conv2(25)	(12, 12, 25)	1525
5	Maxpool2	(6, 6, 25)	0
6	FC5	(5,)	4505